

Introduction to Microeconomics

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SEMINAR #4

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Taxonomy of costs: fixed costs

➤ **Fixed cost (FC)**

- How much does it cost to set up production facilities?
- This is the cost associated with $q = 0$.

➤ DEFINITIONS: the portion of total costs that is invariant in the level of production q .

- FC is equal to zero with constant returns to scale (CRS) technology.

Taxonomy of costs: variable costs

➤ **Variable cost (VC)**

- Varies with q by definition.

➤ DEFINITIONS: the portion of total costs that varies when the level of output q varies.

- VC is proportional to q with CRS technology.

Taxonomy of costs: total costs

➤ **Total cost (TC)**

- Mathematically: $TC(q) = CF + CV(q)$

Taxonomy of costs: average costs

➤ **Average cost (AC)**

- In general, varies with q .
- Mathematically: $AC = TC/q$

➤ DEFINITIONS: how much it costs the firm to produce one unit of output on average.

- AC is constant with CRS technology.

Taxonomy of costs: average variable costs

➤ **Average variable cost (AVC)**

- In general, varies with q .
- Mathematically: $AVC = VC/q$

➤ DEFINITIONS: how much production costs increase on average when the firm increases its output by one unit.

- $AVC = AC$ (constant) with CRS technology.

Taxonomy of costs: marginal costs

➤ **Marginal cost (MC)**

- In general, varies with q .
- Mathematically: $MC = \Delta TC / \Delta q$
- As we will see in next chapter, MC is the key concept when it comes to the firm's choice of output q .
- DEFINITIONS: **how much production costs increase when the firm increases its output by one unit.**
 - $MC = AC = AVC$ with CRS technology.

Exercise 1

The following table shows the evolution of a firm's output together with the number of workers employed:

Workers L	Output q
0	0
1	20
2	50
3	120

$AP = \frac{q}{L}$	$MP = \frac{\Delta q}{\Delta L}$
—	—
20	20
25	30
40	70

Which of the following propositions is correct?

- a) The firm is in a region of decreasing marginal returns. ✗
- b) The average productivity of labor increases then decreases as output increases. ✗
- c) The marginal productivity of labor is increasing for all output levels (from 0 to 120 units). ✓
- d) Marginal productivity equals average productivity for all output levels (from 0 to 120 units). ✗

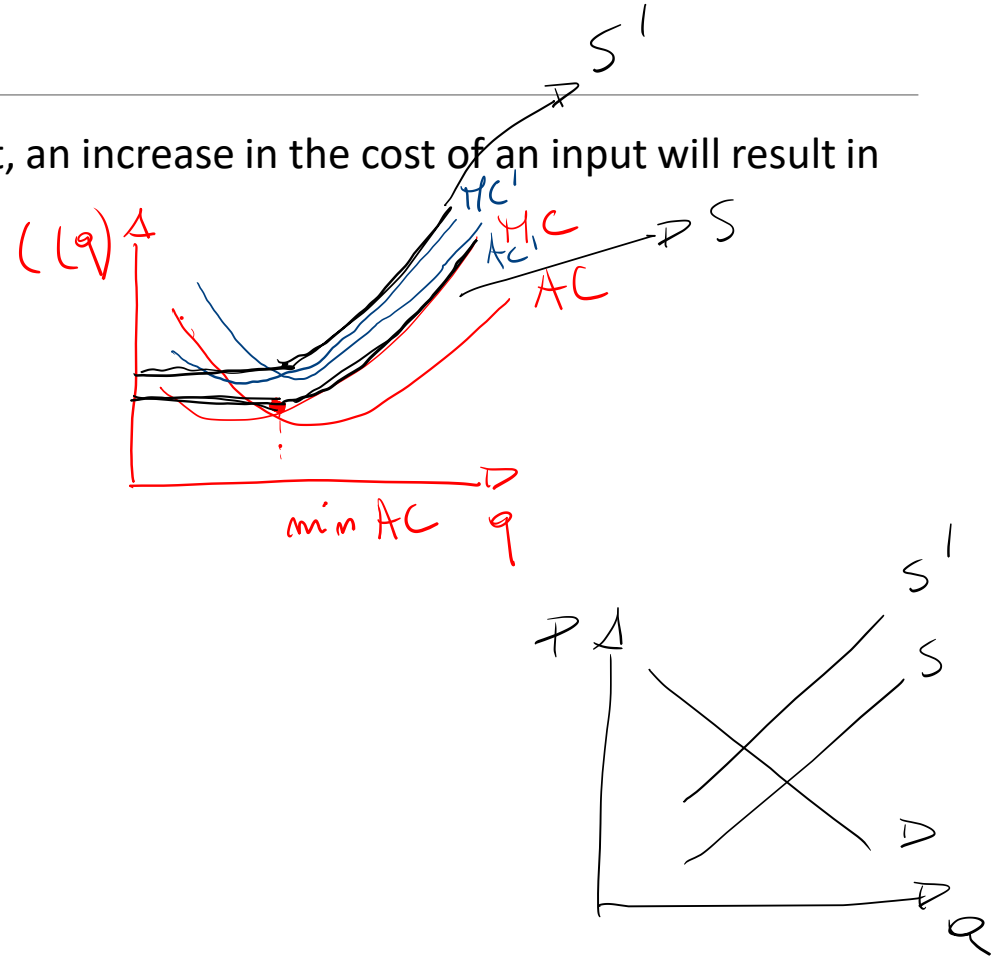
Exercise 2

Consider firms operating in a perfectly competitive environment, an increase in the cost of an input will result in

1. a leftward shift in the firm's supply curve. ✓
2. an upward shift of the firm's marginal cost curve. ✓
3. a leftward shift of the market supply curve. ✓

Which of the following statements are correct?

- a) Only proposition 1 is correct.
- b) Propositions 1, 2 and 3 are correct.
- c) Only proposition 3 is correct.
- d) Propositions 1 and 3 are correct.



Exercise 3

The marginal cost of a business project is summarized in the table below:

Quantity produced	Average Costs	Marginal Costs
0	-	-
1	60	20
2	45	40
3	47	60
4	53	80

Which of the following statements is correct?

- a) The fixed cost is 40. ✓
- b) For the first two units produced, the returns to scale are decreasing. ✓
- c) We can infer that the production technology exhibits constant returns to scale because the marginal cost grows at a constant rate. ✗
- d) None of the above. ✗

$$Q=1 \rightarrow TC=60 = FC + VC$$

$$\begin{array}{r} \downarrow \\ 60 - \\ \underline{20} \\ 40 \end{array} \quad \begin{array}{r} \parallel \\ VC \\ \parallel \\ 20 \end{array}$$

$$Q=2 \rightarrow TC=90 = 40 + VC$$

$$\begin{array}{r} \downarrow \\ FC \end{array} \quad \begin{array}{r} \downarrow \\ 90 - \\ \underline{40} \\ 50 \end{array}$$

$$Q=3 \rightarrow TC=141$$

$$FC \leftarrow 40 + 101$$

$$Q=4 \rightarrow TC=212$$

$$FC \leftarrow 40 + 172$$

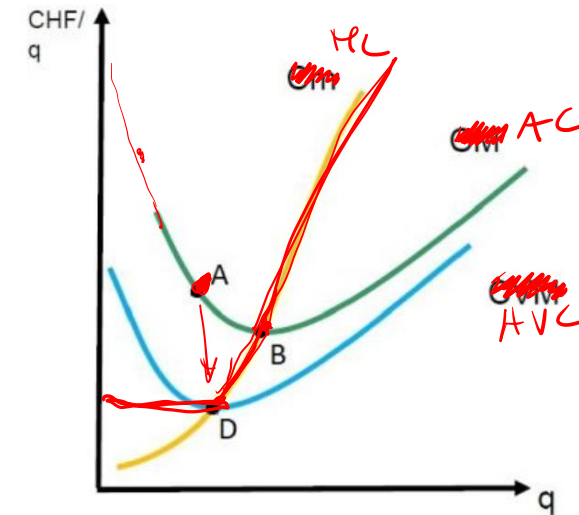
Q	TC	AVC	FC
0	40	—	40
1	60	20	40
2	90	25	40
3	141	33.6	40
4	212	43	40

Exercise 4

$$AC = \frac{TC}{q} = \frac{FC}{q} + \frac{VC}{q} \rightarrow \frac{VC}{q} = AVC$$

$\Rightarrow AC = AVC$

Consider a firm whose marginal cost (MC), average cost (AC) and average variable cost (AVC) functions are represented in the following graph:



Which of the following statements can you state with certainty?

- a) The firm's fixed production costs are zero. ✗
- b) At the level of production corresponding to point A, the last unit produced costs less than the average cost of the units produced so far. ✓
- c) If the long-term level of production corresponds to point B, the firm makes a positive profit. ✗
- d) None of the above prepositions are correct.

Exercise 5

The Ric Levis company produces 62 units of jeans when it employs 4 workers. By hiring an additional worker, production increases to 65 units. What can we say with certainty?

$\frac{\Delta Q}{\Delta L}$
↑

L	Q	AP	MP
4	62	15.5	—
5	65	13	3

- a) The company is in a phase of increasing marginal returns. ✗
- b) The average productivity of labor is less than the marginal productivity of labor. ✗
- c) The marginal productivity of labor is equal to the average productivity of labor. ✗
- d) The average productivity of labor is greater than the marginal productivity of labor. ✓

Exercise 6

In a perfectly competitive industry, the total cost function of a profit-maximizing firm is $TC = 72 + 2Q^2$. If the profit of the firm is zero, then what is the market price?

$$\pi = TR - TC = 0 = P \cdot Q - 72 - 2Q^2$$
$$P = MC = 4Q$$

$$P^* = 4 \cdot Q = 24$$

$$4Q \cdot Q - 72 - 2Q^2 = 0$$

$$\Rightarrow 4Q^2 - 2Q^2 = 72$$

$$Q^2 = 36$$

$$Q^* = 6$$

a) 4

b) 24 ✓

c) 16

d) 64

Exercise 7

$$\begin{aligned}TC(q) &= 16 + q^2 \\Q^D(p) &= 24 - p \\Q^S(p) &= n \cdot q^S(p)\end{aligned}$$

Each company in a competitive market has the following total cost curve: $TC(q) = 16 + q^2$, and market demand is given by: $Q^D(p) = 24 - p$. At the market equilibrium

(Hint: at the market equilibrium, $Q^S(p) = n \cdot q(p)$, where n is the number of companies in the market).

Calculate the price, the quantity produced by each company, the total quantity produced on the market as well as the number of firms present on the market at the long-run equilibrium.

$$\begin{aligned}LR \Rightarrow \pi &= 0 \rightarrow TR = TC \rightarrow Pq = 16 + q^2 \rightarrow 2q \cdot q = 16 + q^2 \\P &= MC = 2q \quad \rightarrow \quad q^2 = 16 \quad \rightarrow \quad q = 4 \\P^* &= 2Q = 8 \\Q^S &= Q^D \\n \cdot q &= 24 - P \\n \cdot 4 &= 24 - 8 \\n &= 16/4 = 4\end{aligned}$$

$$BE \rightarrow \pi = TR - TC = 0 \rightarrow \frac{TR}{q} = \frac{TC}{q} \rightarrow AR = AC$$

$$\begin{array}{c} \parallel \\ MR \\ \parallel \\ MC \\ \parallel \\ P \end{array} \quad MC = AC$$

Exercise 8

close $\Rightarrow MC = AVC$

$$\Rightarrow \frac{q_i}{2} + 8 = \frac{q_i}{4} + 8 \Rightarrow q_i = 0$$

The bicycle market in the city of Amsterdam is said to satisfy the assumptions of perfect competition. There are 10 firms that produce bicycles, each with the following total cost function:

$$TC(q_i) = \underbrace{0.25q_i^2}_{VC} + \underbrace{8q_i}_{\substack{VFC \\ FC}} + 16$$

$$MC(q_i) = \frac{q_i}{2} + 8$$

$$AC(q_i) = \frac{q_i}{4} + 8 + \frac{16}{q_i}$$

With $i = 1, \dots, 10$, q_i the quantity produced by firm i and $TC(q_i)$ the total cost for firm i . The demand in the market is given by the following expression:

$$Q^D = 200 - 10P$$

$$MC(q_i) = AC(q_i)$$

$$\frac{q_i}{2} + 8 = \frac{q_i}{4} + 8 + \frac{16}{q_i}$$

$$\frac{q_i}{4} = \frac{16}{q_i} \rightarrow q_i^2 = 64$$

$$\begin{array}{c} BE \\ q_i = 8 \end{array}$$

Where Q^D represents the quantity demanded and p the price of bicycles.

1. Determine the expression for marginal cost $MC(q_i)$ and average total cost $ATC(q_i)$ of the typical firm.
2. What is the firm's break-even point? What is its closing point?

Exercise 9

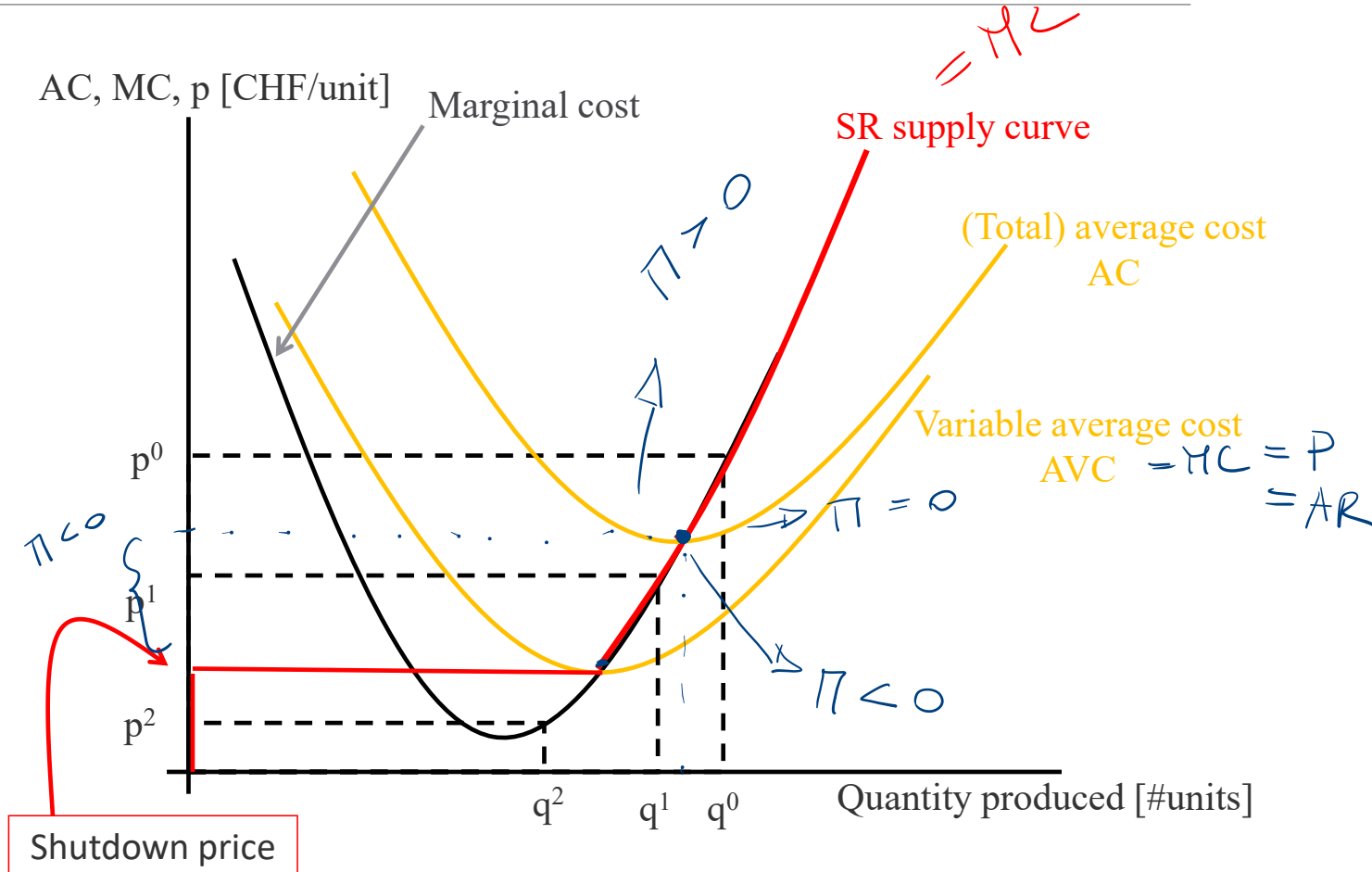
In the case where a firm maximizes its short-run profit by going out of business, which of the following statements is correct?

- a) price < average variable cost for any level of output. ✓
- b) price < marginal cost. ✗
- c) price > minimum of average total cost. ✗
- d) The firm will continue its operations in the short run despite a loss.

$$AVC < P < AC \quad \nearrow \quad \pi = 0$$
$$P < AVC$$

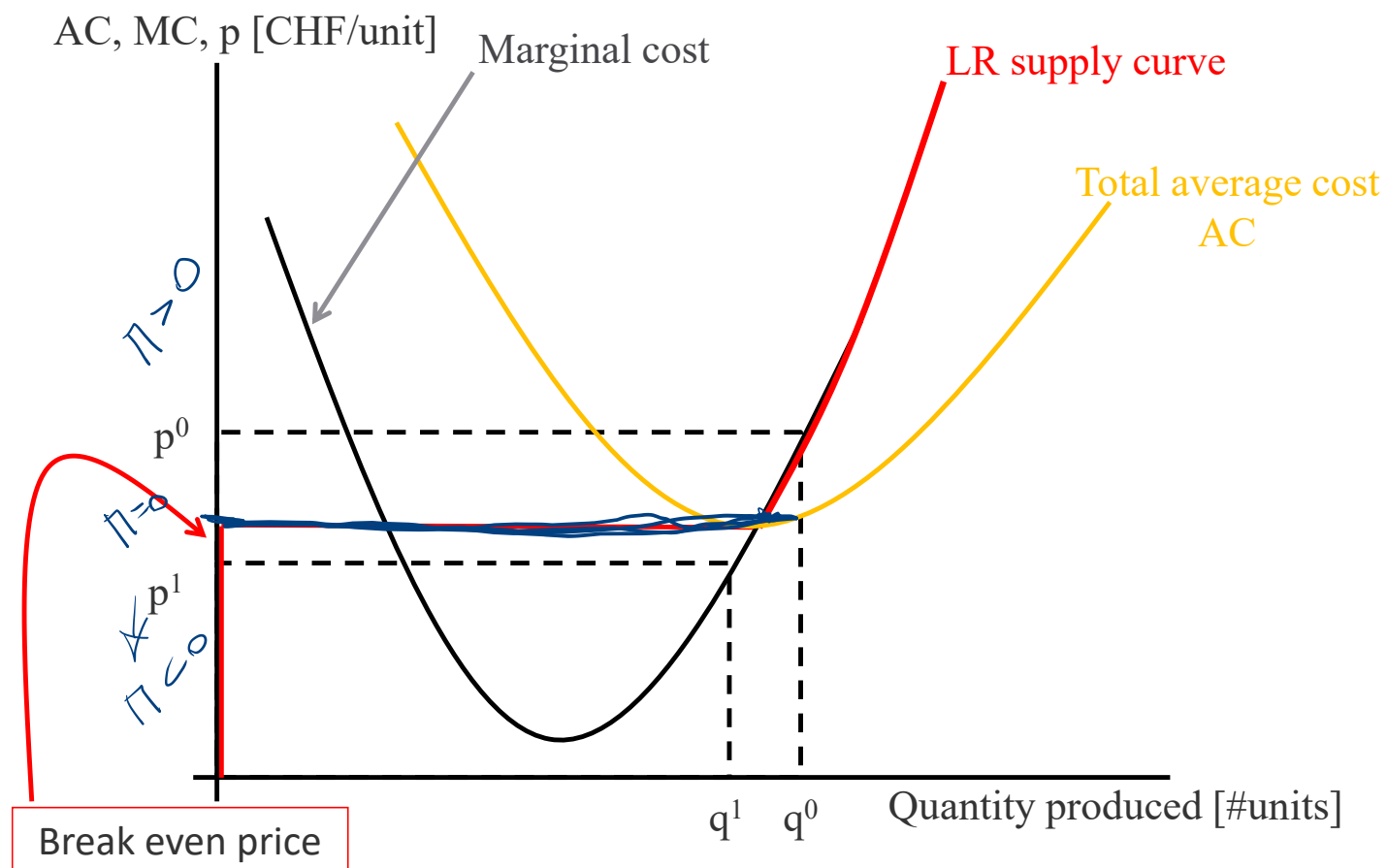
Firm short run supply curve

- At any market price $p < AVC$ (as p^2 in the below graph), the firm shuts down.
- At a price p^1 , the firm will stay on the market, even if it makes losses.

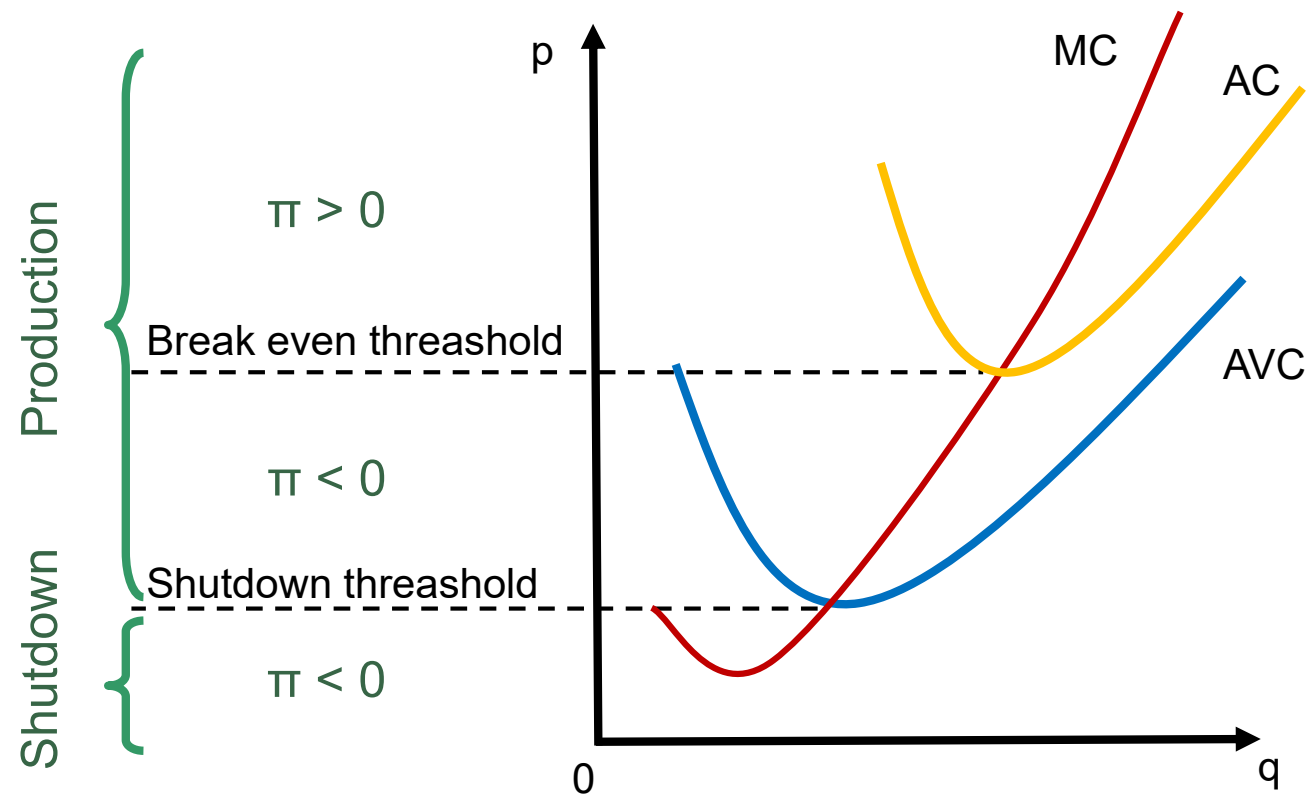


Firm long run supply curve

- At any market price $p < AC$ (as p^1 in the below graph), the firm exits the market.



Firm supply curve: summary

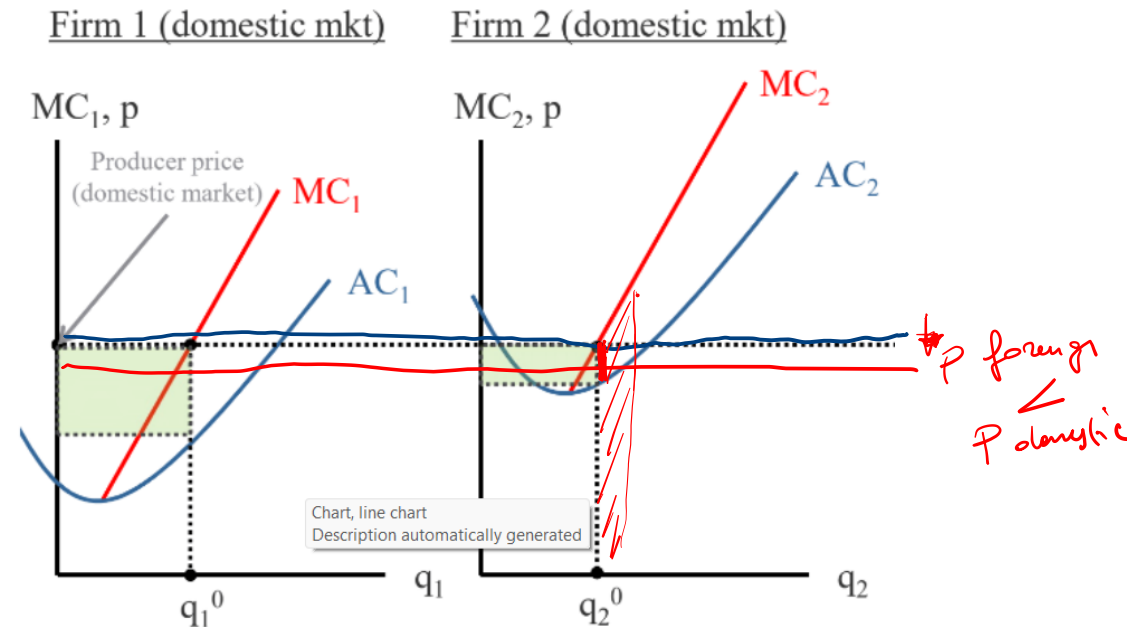


Exercise 10

Consider two heterogeneous firms, with one (firm 1) being more productive than the other (firm 2). Both have duty-free access to their domestic market. They also have the possibility to export their products to a foreign market; if they do so, they face trade costs (which include transportation costs and border levies) so that the producer price net of trade costs is strictly below the consumer price in export markets. Assume that the equilibrium consumer price is identical between the two markets. The figure below depicts marginal and average costs for both firms in the domestic market:

Which of these statements is FALSE?

- a) Firm 2 may choose to export to the foreign market only if the producer price, after deduction of the variable trade cost, is above the average cost. ✓
- b) Firm 1 sells more than firm 2 in the domestic market. ✓
- c) Firm 2 will never choose to export. ✗
- d) Firm 1 makes higher profit, regardless of the decision of firm 2 to export to the foreign market or not. ✓



Exercise 11

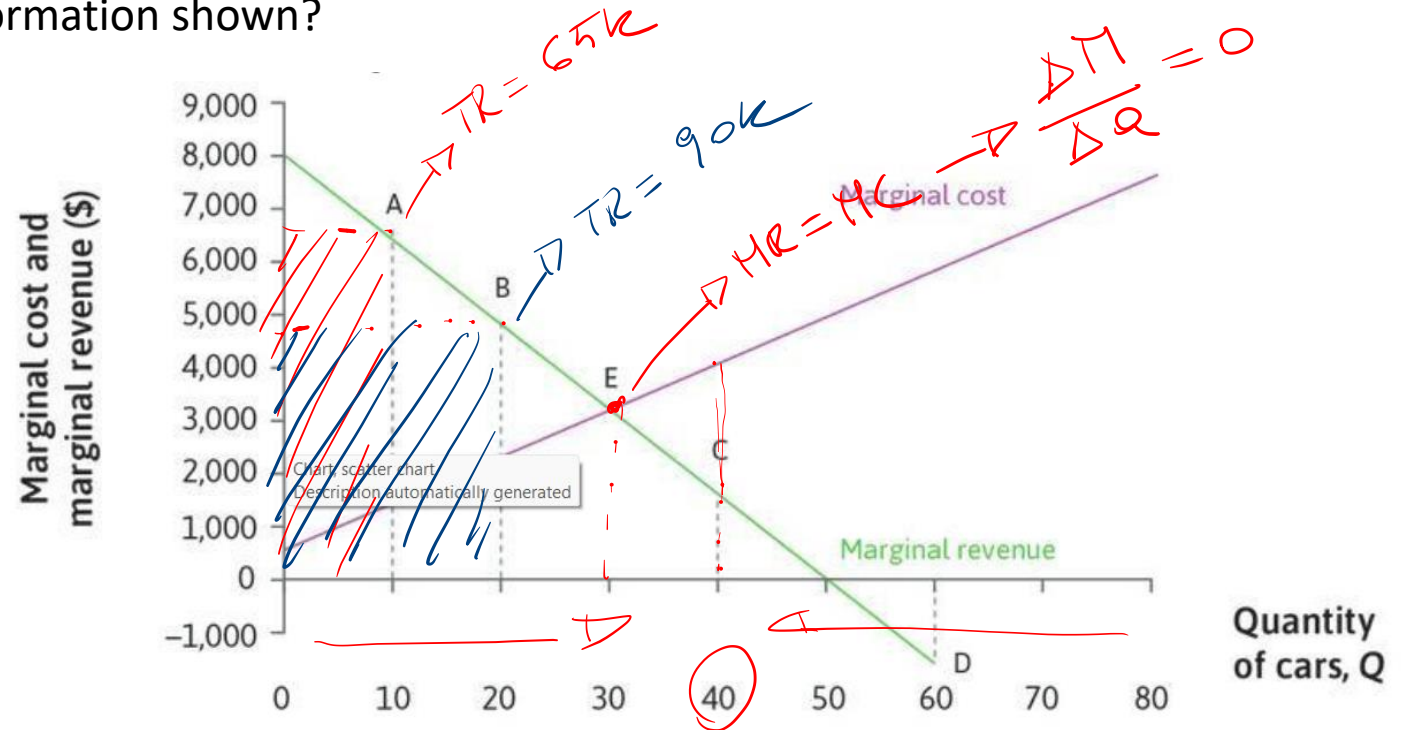
$$\pi = TR - TC$$

$$\frac{\Delta \pi}{\Delta Q} = MR - MC$$

$$\frac{\Delta \pi}{\Delta Q} < 0 \Rightarrow \pi < 0$$

This figure shows the marginal cost and marginal revenue curves for Beautiful Cars. Which of the following statements is correct, based on the information shown?

- a) When $Q = 40$, the marginal cost is greater than the marginal revenue so the firm's profit must be negative. ✗
- b) Revenue is greater when $Q = 10$ than if $Q = 20$. ✗
- c) The firm would not choose to produce at point E because marginal profit is zero. ✗
- d) Profit is greater when $Q = 20$ than when $Q = 10$. ✓



$$P=2 \rightarrow Q^S = 10 \cdot q^A + 8 \cdot q^B = 10 \cdot 20 + 8 \cdot 15 = 320$$

$$P=3 \rightarrow Q^S = 10 \cdot 35 + 8 \cdot 20 = 510$$

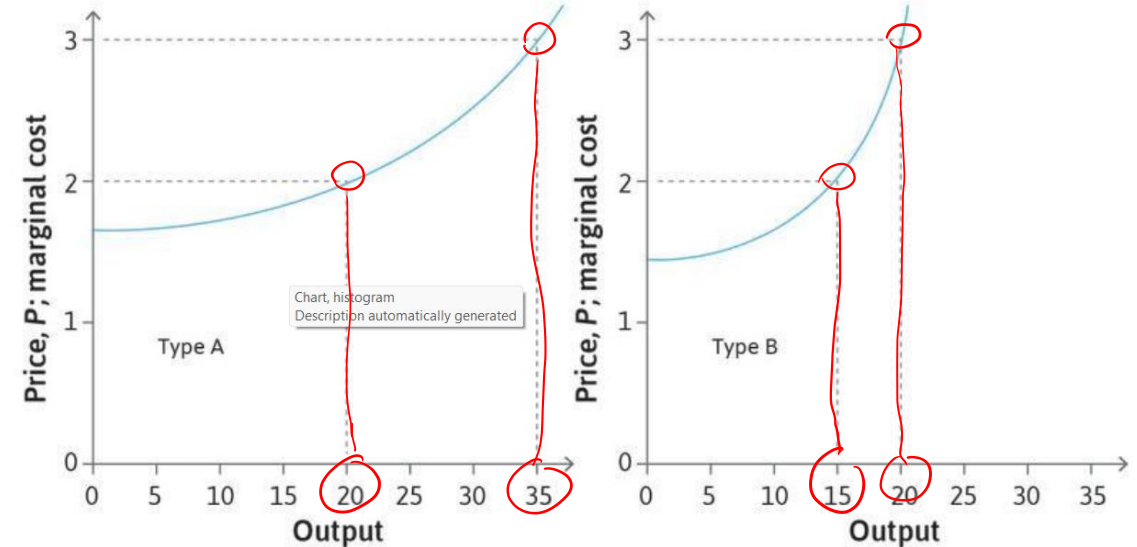
Exercise 12

There are two different types of producers of a good in an industry where firms are price-takers. The marginal cost curves of the two types are given below:

Type A is more efficient than Type B: for example, as shown, at the output of 20 units, the Type A firms have a marginal cost of \$2, as opposed to a marginal cost of \$3 for the Type B firms. There are 10 Type A firms and 8 Type B firms in the market.

Which of the following statements is correct?

- a) At price \$2, the market supply is 450 units. ✗
- b) The market will supply 510 units at price \$3. ✓
- c) At price \$2, the market's marginal cost of supplying one extra unit of the good will depend on the type of the firm that produces it. ✗
- d) With different types of firms, we cannot determine the marginal cost curve for the market. ✗



Exercise 13

Consider a perfectly competitive market. The total cost of a firm is $TC = 500 + 0.25Q^2$. If the profit-maximizing firm faces $P = 50$, then firm's profit will be:

a) 2000 ✓

b) 1000

c) 2500

d) 1500

$$P = MC = \frac{Q}{2} = 50 \rightarrow Q^* = 100$$

$$\begin{aligned}\Pi &= TR - TC = P \cdot Q - 500 - \frac{Q^2}{4} \\ &= 50 \cdot 100 - 500 - \frac{10000}{4} \\ &= 5000 - 3000 = 2000\end{aligned}$$

Exercise 14

Suppose the total cost of a firm is given by $TC = 50 + q^2$. If there are 200 identical firms in a perfectly competitive market, then the market supply curve is:

a) $Q = 50P$

b) $Q = 400P$

c) $Q = 150P$

d) $Q = 100P$ ✓

$$MC = P = 2Q$$

$$Q = \frac{P}{2}$$

$$Q^s = 200 \cdot Q = \frac{200}{2} P = 100P$$

Exercise 15

$$MC = Q = P = 4$$

The total cost of a firm is given by $TC = 10 + 0.5q^2$. Which of the following statements is correct?

a) The marginal cost is constant. $\times$

b) In perfect competition, with a market price of 4 CHF, the total cost associated to profit-maximizing quantity is 18 CHF. $\checkmark$

c) For all output levels, the average cost is decreasing. $\times$

d) None of the above propositions is correct. $\times$

$$q = 4 \rightarrow TC = 10 + 8 = 18$$

$$+ \frac{1}{2} \frac{4^2}{2}$$

$$AC = \frac{10}{q} + \frac{q}{2}$$

